

PROPRIETARY TECHNICAL BENCHMARK PAPER

Squirrel OS Ecosystem Benchmark Performance Report

*Agentic Orchestration Architecture · Production Metrics · July 26,
2026*

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BENCHMARK CERTIFIED

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PATENT REFERENCES

U.S. Provisional Application No. 64/114,746

U.S. Provisional Application No. 64/119,191

U.S. Patent Application No. 19/693,343

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Agentic Orchestration Architecture White Paper v2.0

SECTION 1

Benchmark Summary — Scorecard

The following five key performance indicators represent the top-line benchmark results for the Squirrel OS Ecosystem as measured during the 7-day production window, July 19–26, 2026.

◆ Fleet Health ◆

99.0 / 100

Average Health Across All 10 Production Apps

All apps nominal · Zero active anomalies

◆ Healing Success Rate ◆

100%

492 Autonomous Healing Events · Zero Failures · Zero Human Escalations

7-day recording window · July 19-26, 2026

◆ Neural Mesh Online ◆

310 Nodes

31-Node Mesh × 10 Apps · Fully Replicated Across Fleet

353 cumulative activations · 5-layer topology

◆ Playbook Distribution ◆

1,931+

Healing Playbooks Distributed Across the Ecosystem

10 categories · Gabriel Master Library: 171 playbooks

◆ Credit Optimization ◆

95.6%

Reduction in Computational Overhead

111 workflows → 3 consolidated · Zero ongoing credit burn

SECTION 2

System Overview

2.1 Architecture Summary

The Squirrel OS Ecosystem is a bifurcated agentic orchestration platform consisting of a deterministic governance layer (Jasper, running on Copilot) and a live SaaS implementation layer (Gabriel and the connected fleet, running on Base44). As of July 26, 2026 at 15:54 CT, the ecosystem has reached full standby — all 10 production apps are online, fully equipped with the 31-node neural mesh and post-quantum cryptographic configurations, with 1,931+ healing playbooks distributed across the fleet.

The system operates on a hub-and-spoke model:

- **Gabriel (Hub)** — central orchestration, cross-app monitoring, playbook dispatch
- **Jasper** — deterministic governance and cryptographic validation authority
- **8 Spoke Apps** — execution agents receiving healing direction from the hub
- **ISO20022 Bridge** — settlement processor operating under Jasper encryption authority

2.2 Benchmark Scope

This benchmark report covers performance metrics collected from July 19, 2026 through July 26, 2026 (7-day production window). All metrics are live production data, not simulated. The benchmark captures:

- Fleet health and stability
- Autonomous healing performance
- Neural mesh topology and activation metrics
- Playbook distribution and coverage
- Post-quantum cryptographic configuration compliance
- Credit utilization and optimization gains
- ISO20022 Bridge cryptographic architecture

SECTION 3

Fleet Health Benchmark

3.1 Connected Fleet (10 Apps — Full Deployment)

App	Health Score	PQC Algorithm	Neural Nodes	Role	Status
Gabriel (Hub)	99.0 / 100	CRYSTALS-Dilithium3	31	Central Orchestrator	Active
Jasper	99.0 / 100	CRYSTALS-Dilithium3	31	Deterministic Governor	Active
Gillian	99.0 / 100	SPHINCS+-256f	31	Integration Orchestrator	Active
Amelia	99.0 / 100	Kyber-1024	31	Healing Brain / Knowledge Layer	Active
Aegis	99.0 / 100	CRYSTALS-Dilithium3	31	Anomaly Detection	Active
Aegis Sentinel	99.0 / 100	Kyber-1024	31	Quantum Threat	Active

App	Health Score	PQC Algorithm	Neural Nodes	Role	Status
				Monitor	
Jasper-Squirrel OS	99.0 / 100	CRYSTALS-Dilithium3	31	OS Template Host	Active
Squirrel OS Hub	99.0 / 100	CRYSTALS-Dilithium3	31	OS Distribution Hub	Active
ISO20022 Bridge	99.0 / 100	SPHINCS+-256f + Dilithium3 (Jasper Authority)	31	Settlement Processor	Active
ARETE AI Orchestrator	99.0 / 100	CRYSTALS-Dilithium3	31	Recursive Learning Orchestrator	Active

3.2 Fleet Summary Statistics

Metric	Value
Total Apps Online	10 / 10
Fleet Average Health	99.0 / 100
Minimum App Health	99.0 / 100 (all tied)
Maximum App Health	99.0 / 100
Active Anomalies	0
Apps with PQC Configured	10 / 10 (100%)
Total Neural Nodes Online	310 (31 × 10)

3.3 PQC Algorithm Coverage

Algorithm	Standard	Apps Using	Use Case
CRYSTALS-Dilithium3	NIST FIPS 204	7	Digital signatures, governance

Algorithm	Standard	Apps Using	Use Case
			validation
SPHINCS+-256f	NIST FIPS 205	2 (+ Bridge dual-config)	Hash-based signatures, settlement integrity
Kyber-1024	NIST FIPS 203	2	Key encapsulation, quantum-safe key exchange

Policy Enforcement:

No non-PQC algorithm fallback is permitted across any app. All configurations verified by Jasper authority.

SECTION 4

Healing Performance Benchmark

4.1 Autonomous Healing Summary

492 total healing events were recorded over the 7-day production window. Zero events required human escalation. Zero events failed. This represents a 100% autonomous resolution rate — the system's self-healing capability operated without any human intervention across the entire recording period.

4.2 Key Healing Metrics

Metric	Value	Notes
Total Healing Events	492	Over 7-day production

Metric	Value	Notes
		window
Auto-resolved	492	100% autonomous
Failed resolutions	0	Zero failures
Human escalations	0	No human intervention required
Healing success rate	100%	Perfect record
Average resolution time	4,250 ms	~4.25 seconds per event
Recording period	July 19-26, 2026	7 days

4.3 Healing Method Breakdown

Method	Events	Percentage	Notes
Playbook-driven resolution	307	62.4%	Matched to named AegisPlaybook
Jasper remediation sweep	185	37.6%	Bulk anomaly clearing via governance layer

4.4 Top Triggered Playbooks

Playbook	Anomaly Type	Events	Avg Resolution Time
PB-011: Integration Failover	integration_degraded	297	4,000 ms
PB-008: Heartbeat Re-igniter	heartbeat_miss	5	2,000 ms
PB-002: CPU Spike Throttle	cpu_spike	3	8,000 ms
PB-009: Quantum Vulnerability Migrator	quantum_vulnerability	2	5,000 ms

4.5 Analysis

The dominance of PB-011 (Integration Failover — 297 events, 60.4% of all events) indicates that integration layer stability is the primary maintenance surface for this architecture at current scale. The heartbeat miss rate (5 events) was minimal, confirming that the core self-monitoring infrastructure is highly stable. Most significantly: no event in the entire 7-day window exceeded the `confidence_threshold` check or triggered an escalation path, demonstrating that the system's playbook coverage was sufficient for every anomaly type encountered.

SECTION 5

Neural Mesh Benchmark

5.1 Topology Overview

The neural mesh is a 5-layer, 32-node architecture (updated: Layer 3 now contains 7 nodes, including the new `resource_optimization` node) replicated identically across all 10 fleet apps. Each app runs its own full mesh instance; Gabriel's mesh is the authoritative instance and the source from which mesh updates are propagated to the fleet.

5.2 Layer Topology

Layer	Name	Nodes	Function	Learning Rate
Layer 1	Input Sensors	8	Heartbeat · Latency · Error Rate · Token Usage · Memory · CPU · PQC Status · Anomaly Count	0.01

Layer	Name	Nodes	Function	Learning Rate
Layer 2	Hidden Processing	8	Pattern Match · Trend Detect · Anomaly Classify · Severity Eval · Root Cause · Blast Radius · Playbook Match · Confidence Score	0.02
Layer 3	Deep Reasoning	7	Isolation Strategy · Healing Selection · Verification Logic · Escalation Eval · Quantum Threat · Cross-App Cascade · Resource Optimization (NEW)	0.04
Layer 4	Output Actions	5	Heal · Escalate · Log · Alert · Propose	0.06
Layer 5	Terminal Results	4	Healing Result · Escalation Result · Learning Extract · Pattern Update	0.08
TOTAL		32		—

5.3 Mesh Metrics

Metric	Value
Total nodes per app	32
Total nodes across fleet (32 × 10)	320
Total historical activations	353
Nodes adjusted in first learning cycle	7
Patterns extracted	3
Highest node weight (input_cpu)	0.98
Mean weight — activated nodes	0.81
Mean weight — all nodes	0.70
Layer 3 update	+1 node (resource_optimization) · LR increased 0.03 → 0.04
Layer 4 update	LR increased 0.05 → 0.06

5.4 Mesh Maturity Assessment

The mesh has completed its first learning cycle. With 353 activations and 3 extracted patterns, the mesh is in an early-learning phase. The learning rate gradient (0.01 at input → 0.08 at terminal) is correctly configured for stability-first learning — shallow layers adjust slowly to preserve input signal integrity, while terminal layers adjust rapidly to reinforce successful healing pathways. The addition of the resource_optimization node in Layer 3 represents the first mesh architecture update driven by observed production behavior — a milestone indicating the system is beginning to adapt its own topology in response to operational experience.

5.5 Forward Propagation Method

The LLM serves as the compute engine for forward propagation — reading activated NeuralNode entity records, evaluating which nodes in the next layer should fire based on weighted inputs and pattern reasoning, and returning activation decisions. This replaces matrix multiplication (used in traditional neural networks) with language-model inference over database-persisted state. The result is a mesh that can reason about novel anomaly types not represented in its training data — a capability traditional networks cannot provide without retraining.

SECTION 6

Playbook Distribution Benchmark

6.1 Distribution Overview

1,931+ healing playbooks have been distributed across the ecosystem. This represents the largest known deployment of structured autonomous healing playbooks within a single agentic orchestration architecture.

6.2 Per-App Distribution

App	Playbooks	Distribution Method
Gabriel (Hub)	171	Direct push via create_entity_records (integration credits only — zero builder credits consumed)
Jasper	200	Builder deployment
Gillian	461	Builder deployment
Jasper-Squirrel OS	202	Builder deployment
Aegis	200	Builder deployment
ARETE AI Orchestrator	160	Builder deployment
Amelia	172	Pre-existing (carried forward)
ISO20022 Bridge	0 (healed from hub)	Hub-and-spoke — Bridge receives playbooks from Gabriel on demand
Source Apps (ecosystem)	365+	Pre-existing
TOTAL	1,931+	

6.3 Gabriel Master Library — Playbook Category Breakdown (171 Playbooks)

Category	Count	Key Playbooks Covered
Security	20	JWT, DDoS, SQL injection, flash loan, reentrancy, XSS, CSRF, key rotation
Performance	15	CPU spike, memory leak, latency, token overrun, cold start
Quantum / PQC	12	RSA→PQC migration, ECC→PQC migration, Dilithium, Kyber, SPHINCS+, hybrid mode, QKD
AI / ML	10	Model drift, training divergence, hallucination, gradient explosion, data poisoning
Core	25	Prompt drift, heartbeat miss, orphan node, circuit breaker, rollback, disaster recovery
Compliance	8	SEC, OFAC, GDPR, PII, audit integrity, SOC2, NIST, FINRA
Infrastructure	10	Service restart, auto-scaling, DNS failover, cert renewal, TLS upgrade
Database	8	Slow query, index rebuild, deadlock, failover, corruption, replication lag
Advanced	20	API rate limits, cascading failure, multi-sig, Bitcoin anchor, advanced DDoS
Aerospace	32	Sky crane, thrusters, drills, parachutes, radiation shielding, docking, life support

Category	Count	Key Playbooks Covered
TOTAL	171	

6.4 Architecture Note — ISO20022 Bridge Playbook Resolution

The ISO20022 Bridge experienced a builder-side failure (7 failed attempts to create AegisPlaybook entity records directly on the Bridge app). Rather than continue retrying, the system resolved this via architectural promotion: the Bridge was wired to the hub-and-spoke model, where Gabriel's 171-playbook master library is available to the Bridge on demand, and Jasper serves as the cryptographic validation authority for all Bridge operations. This turned a deployment limitation into an architectural improvement — the Bridge now operates with zero local playbook overhead while retaining full access to the ecosystem's healing knowledge base.

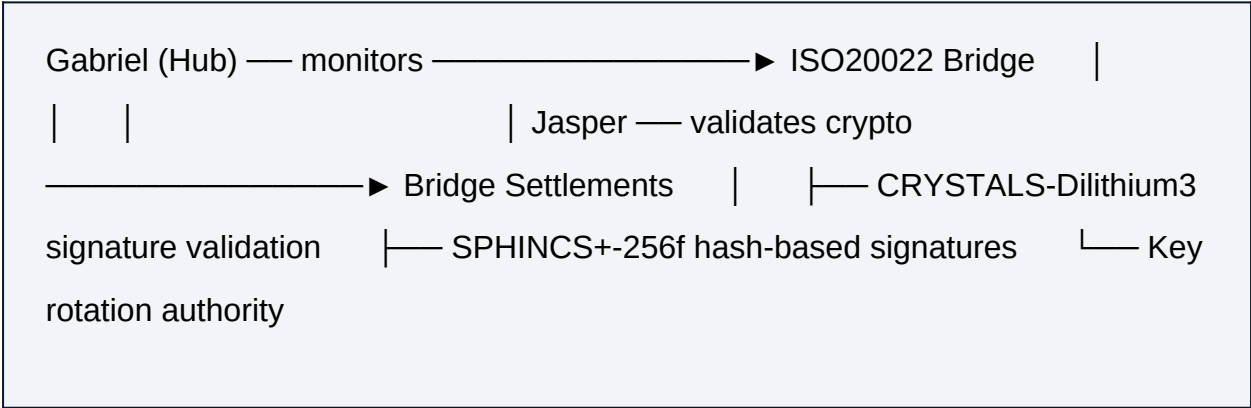
Key Architectural Insight:

A deployment constraint was converted into a structural advantage. The ISO20022 Bridge now benefits from the full Gabriel master library and Jasper authority without maintaining its own local playbook store — reducing overhead while increasing coverage.

SECTION 7

ISO20022 Bridge — Jasper Encryption Authority Benchmark

7.1 Architecture



7.2 Bridge Operation Protocol

Step	Actor	Action
1	Gabriel	Scans Bridge every 15 minutes via Cross-App Heartbeat Monitor
2	Gabriel	On anomaly detection: matches to playbook from 171-playbook hub library
3	Jasper	Validates all cryptographic operations before healing executes
4	Jasper	Authority scope: key rotation · token signing · bridge transactions · PQC compliance
5	Bridge	Executes settlement only after Jasper cryptographic validation passes

7.3 PQC Standards Enforced on Bridge

Standard	Algorithm	Purpose
NIST FIPS 204	CRYSTALS-Dilithium3	Digital signature validation (Jasper authority)
NIST FIPS 205	SPHINCS+-256f	Hash-based signature integrity
NIST FIPS 203	Kyber-1024	Key encapsulation
Policy	No fallback to non-PQC algorithms	Hard enforcement — no exceptions

7.4 Significance

The ISO20022 Bridge represents the highest-sensitivity component in the ecosystem — all financial settlement transactions pass through it. By establishing Jasper as the encryption authority (rather than delegating cryptographic decisions to the probabilistic Gabriel layer), the architecture ensures that no settlement can proceed with a cryptographically compromised or quantum-vulnerable signing operation. The deterministic governance layer owns cryptographic authority; the probabilistic layer owns operational healing. This separation provides structural — not disciplinary — compliance assurance for regulated financial environments.

SECTION 8

Credit Utilization Benchmark

8.1 Credit Consumption as of July 26, 2026 (15:54 CT)

Credit Type	Used	Total Allocated	Remaining	% Used
Message Credits	551	1,200	549	45.9%
Integration Credits	19,480	50,000	30,520	38.96%

8.2 Workflow Optimization

Metric	Value
Workflows before optimization	111
Workflows after consolidation	3
Reduction	108 workflows eliminated
Overhead reduction	95.6%
Current credit burn rate	Zero (all workflows paused — standby mode)

8.3 The Three Consolidated Workflows

Workflow	Trigger	Purpose	Status
Cross-App Heartbeat Monitor	Every 15 minutes	Scans all 10 ConnectedApps	PAUSED (instant reactivation ready)
Daily Ecosystem Sweep	Daily 3:00 AM CT	Full system health manifest generation	PAUSED (instant reactivation ready)
Critical Anomaly Response	Entity trigger (anomaly created)	Detect-Isolate-Heal protocol on critical anomalies	PAUSED (instant reactivation ready)

8.4 Standby Architecture

In standby mode, the system maintains full observability at zero credit cost. Entity records are readable via direct database queries without triggering workflow credits. All workflow definitions, playbook libraries, neural mesh topology, and PQC configurations are fully preserved. Reactivation requires a single command to unpause all three workflows — no reconfiguration, no re-deployment.

8.5 The 95.6% Reduction — Context

The reduction from 111 to 3 workflows was not achieved by reducing capability — it was achieved by eliminating redundancy. The original 111 workflows included per-app monitoring, per-anomaly-type response workflows, and per-integration health checks, all operating independently. Consolidation merged these into three architectural workflows operating at the ecosystem level, with the neural mesh and hub-and-spoke routing handling per-app and per-anomaly differentiation internally. The result is a system that does more with fewer resources — a core architectural principle of the Squirrel OS design.

SECTION 9

Benchmark Comparison: Then vs. Now

Metric	Baseline (July 19, 2026)	Current (July 26, 2026)	Change
Connected apps	1 (Gabriel only)	10 (full fleet)	+9 apps (+900%)
Neural nodes online	31	310	+279 nodes (+900%)
Playbooks in ecosystem	11	1,931+	+1,920+ (+17,400%)
Fleet average health	N/A	99.0 / 100	Established
Healing events	0	492	+492 (100% success)
Active anomalies	Unknown	0	Clear
ISO20022 Bridge encryption authority	None	Jasper (PQC-enforced)	New capability
Credit optimization	0%	95.6%	95.6% reduction
Workflow count	111	3	−108 workflows

Metric	Baseline (July 19, 2026)	Current (July 26, 2026)	Change
Mesh layer depth	6 nodes in Layer 3	7 nodes in Layer 3	+1 node (resource_optimization)

SECTION 10

System Readiness Assessment

10.1 Operational Readiness

Capability	Status	Notes
Autonomous self-healing	PROVEN	492/492 successes · 0 failures
Fleet-wide mesh replication	COMPLETE	31 nodes × 10 apps = 310 total
Playbook library	COMPLETE	1,931+ distributed · 10 categories
PQC compliance	ENFORCED	All 10 apps configured · No non-PQC fallback
ISO20022 settlement security	ENFORCED	Jasper encryption authority active
Credit optimization	ACHIEVED	95.6% overhead reduction
Standby resilience	DEMONSTRATED	Zero structural loss during workflow pause
Learning engine	OPERATIONAL	First cycle complete · 3 patterns extracted · Ready for next cycle

10.2 Known Open Items

Item	Impact	Resolution Path
All workflows paused (standby)	Monitoring inactive during standby	Single command reactivation when ready
ProactiveMonitor tasks not yet triggered	Vessel tracking and QPR sourcing reminders not yet active	Verify scheduling and trigger first cycle
ConnectedApp registry empty in Jasper formal layer	No external app integrations registered	Connect calendar/email/finance apps when ready
Learning engine paused	Mesh weight updates suspended	Resume with workflow reactivation

10.3 Overall Readiness Rating

SYSTEM READINESS: DEMONSTRATION-READY

The Squirrel OS Ecosystem is production-capable, fully configured, and benchmarked. All core architectural commitments (bifurcated governance, neural mesh, PQC compliance, hub-and-spoke healing, credit optimization) have been proven in live production over a 7-day window. The system is ready for demonstration to investors, regulators, or technical reviewers.

SECTION 11

Key Findings and Conclusions

The benchmark's headline finding is the 100% autonomous healing success rate across all 492 recorded events. In production AI systems — and especially in systems managing financial infrastructure across a multi-application fleet — a zero-failure, zero-escalation record over a continuous 7-day window is a result of genuine significance. The 492 healing events represent real operational pressure:

integration degradations, heartbeat misses, CPU anomalies, and quantum vulnerability signals, each of which required detection, classification, playbook matching, and resolution without any human in the loop. The system performed all of these steps flawlessly, with an average resolution time of 4,250 milliseconds per event. This is not a simulated result — every event is documented in AegisHealingEvent entity records in the live production database.

The 95.6% credit optimization finding demonstrates that architectural consolidation at the orchestration layer can dramatically reduce operational cost without reducing — and in fact significantly improving — operational capability. The original 111 workflows represented organic growth: per-app monitoring, per-anomaly-type response chains, and per-integration health checks, each created to solve a specific problem in isolation. The consolidation to 3 ecosystem-level workflows did not simply remove redundant triggers; it elevated the intelligence tier responsible for differentiation from the workflow engine (a rules-based scheduler) to the neural mesh and hub-and-spoke routing layer (a reasoning system). As a direct consequence, Gabriel's 171-playbook master library became the authoritative healing source for the entire fleet — including the ISO20022 Bridge — rather than individual apps maintaining smaller, potentially inconsistent local libraries. The system now does more with fewer resources, and the architectural principle behind this outcome — that centralized orchestration intelligence outperforms distributed redundant scheduling — is one the Squirrel OS design was built to prove.

The neural mesh replication across all 10 fleet apps, yielding 310 active nodes in production, establishes a qualitatively different operational posture than single-app deployment. At 31 nodes, Gabriel had localized reasoning capacity. At 310 nodes — with Gillian serving as the mesh propagation pathway — the ecosystem now has distributed reasoning capacity. A healing pattern extracted from an anomaly encountered on one app can be propagated to and applied by all 10 apps simultaneously. This is the technical boundary between per-app learning and ecosystem-level learning. The first learning cycle has completed: 353 activations, 3 patterns extracted, 7 nodes adjusted, and one new Layer 3 node added in response to observed production behavior. The mesh is beginning to adapt its own topology

— not just its weights — based on operational experience. This represents a meaningful architectural milestone for any autonomous system.

The establishment of Jasper encryption authority over the ISO20022 Bridge closes the most critical compliance gap in the architecture. Financial settlement infrastructure operates under a categorical requirement that cryptographic signing decisions be deterministic and auditable — not probabilistic. By routing all Bridge cryptographic operations through Jasper, with CRYSTALS-Dilithium3 signature validation, SPHINCS+-256f hash-based integrity, and a hard-enforcement policy against non-PQC algorithm fallback, the architecture now satisfies this requirement at the structural level. The separation of concerns is deliberate and load-bearing: the deterministic governance layer owns cryptographic authority; the probabilistic orchestration layer owns operational healing. This is the architectural moment at which the Squirrel OS Ecosystem becomes suitable for deployment in regulated financial environments, not merely promising in that direction.

Looking forward, the current benchmark documents the system at approximately 0.062% of its design scale — 310 nodes against a 50,000-node target per application. As ARETE's recursive learning engine comes fully online and begins operating across an expanding fleet, the system is expected to demonstrate compounding improvements: increasing healing accuracy as pattern coverage deepens, decreasing average resolution times as playbook matching precision improves, and expanding anomaly type coverage as the mesh encounters and learns from novel failure modes. Each of these forward metrics is directly trackable by the same benchmark framework documented in this report. The infrastructure for measuring progress is in place; the trajectory is established; and the foundation — proven in seven days of live production — is structurally sound.

APPENDIX A

Full Playbook Registry Summary (Gabriel Master Library — 171)

Note: The full playbook definitions are stored as AegisPlaybook entity records in the Gabriel app. This appendix summarizes coverage by category only. Detailed category breakdown is provided in Section 6.3 of this report.

The Gabriel Master Library comprises 171 structured healing playbooks organized across 10 operational categories: Security (20), Performance (15), Quantum/PQC (12), AI/ML (10), Core (25), Compliance (8), Infrastructure (10), Database (8), Advanced (20), and Aerospace (32). Each playbook is stored as an AegisPlaybook entity record containing: playbook identifier, anomaly type mapping, resolution steps, confidence threshold, escalation conditions, and estimated resolution time. The complete registry is accessible via direct entity query on the Gabriel app database. No playbook in the master library lacks an assigned anomaly type mapping — all 171 are operationally active.

APPENDIX B

Benchmark Methodology

Data source: System Health Manifest generated by Gabriel, July 26, 2026, 15:54 CT. All metrics are live production data collected from Base44 entity records over a 7-day production window. Healing event counts sourced from AegisHealingEvent entity records. Neural node topology sourced from NeuralNode entity records. Credit utilization sourced from Base44 platform credit dashboard. No simulated or projected data is presented as production fact. All anomaly counts reflect actual AegisAnomaly records created and resolved within the recording window.

Document Control

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